



Introduction

Advances in spatial epidemiology and geographic information systems

Russell S. Kirby PhD, MS^{a,*}, Eric Delmelle PhD^b, Jan M. Eberth PhD, MSPH^c^a Department of Community and Family Health, University of South Florida, Tampa^b Department of Geography and Earth Sciences, University of North Carolina-Charlotte, Charlotte^c Department of Epidemiology and Biostatistics, University of South Carolina, Columbia

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ABSTRACT

The field of spatial epidemiology has evolved rapidly in the past 2 decades. This study serves as a brief introduction to spatial epidemiology and the use of geographic information systems in applied research in epidemiology. We highlight technical developments and highlight opportunities to apply spatial analytic methods in epidemiologic research, focusing on methodologies involving geocoding, distance estimation, residential mobility, record linkage and data integration, spatial and spatio-temporal clustering, small area estimation, and Bayesian applications to disease mapping. The articles included in this issue incorporate many of these methods into their study designs and analytical frameworks. It is our hope that these studies will spur further development and utilization of spatial analysis and geographic information systems in epidemiologic research.

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Introduction

Defining spatial epidemiology

All students of epidemiology learn that descriptive epidemiology focuses on the triad of person, place, and time [1]. Although epidemiologic research focusing on place or location historically received considerably less attention, modern epidemiology increasingly incorporates the spatial perspective into research designs and models. Spatial factors have also become prominent features in etiologic research, especially concerning host-vector-agent interactions, but also in guiding social and environmental epidemiologic investigations. Spatial methods are also progressively incorporated into health services research focused on specific diseases, health conditions, or risk factors.

The field of spatial epidemiology has evolved over the years. Elliott et al. [2] identified four types of spatial analyses in epidemiology: (1) disease mapping, (2) geographical correlation studies, (3) risk assessment in relation to point or line sources, and (4)

cluster detection and disease clustering. Only a few years earlier, English [3] had defined geographical epidemiology as “the description of spatial patterns of disease incidence and mortality.” More recently, in a widely used spatial statistics text, Lawson [4] states that spatial epidemiology “concerns the analysis of the spatial/geographical distribution of the incidence of disease.” Here, we argue that spatial epidemiology encompasses research that incorporates the spatial perspective into the design and analysis of the distribution, determinants, and outcomes of all aspects of health and well-being across the continuum from prevention to treatment.

Spatial epidemiology is not synonymous with health and/or medical geography. Spatial epidemiology refers to inquiries that use epidemiologic study designs that involve spatial data or spatially derived information about study subjects, health facilities, or sources of exposure. Health or medical geography, a subdiscipline of human geography, encompasses research applying geographic analytical methods to health, disease, or health care issues. Its distinguishing feature is the primary focus on spatial patterns and context, whereas spatial epidemiology is inherently focused on populations [5]. Although many studies, especially those involving interprofessional research teams, combine the methods of both disciplines in creative and innovative ways, far more often spatial epidemiologic research does not fully incorporate the geographic perspective, and vice versa [6].

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* Corresponding author. Department of Community and Family Health, College of Public Health, University of South Florida, 13201 Bruce B. Downs Blvd, MDC56, Tampa, FL 33612. Tel.: +1-813-396-2347; fax: +1-813-905-9998.

E-mail address: rkirby@health.usf.edu (R.S. Kirby).

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In this brief overview, we argue for an expanded role for spatial epidemiology within our discipline and demonstrate the importance of a broader scope for spatial perspectives in the study of epidemiology through the series of articles that follow in this symposium on “Geographic Information Systems and Spatial Methods in Epidemiology.”

The concepts of place and neighborhood

Health outcomes are influenced by an interplay of different factors, such as individual attributes, the physical and social environments an individual interacts with, cultural norms, and both the provision and utilization of health services [7]. It is widely recognized that the place where an individual lives or works should be considered as a potential disease determinant [8]. For example, women living in rural areas may have to travel longer distances to reach mammography facilities, potentially leading to a decrease of mammographic breast screening or a delay in their diagnosis [9–11]. Children living in a pedestrian friendly environment where parks and playgrounds are readily accessible are more likely to engage in physical activity, reducing the odds of obesity [12,13]. Along the same vein, higher risk factors for obesity are generally observed in food desert areas, which are characterized by poor-quality food environments and a lack of supply of supermarkets with fresh food [14,15].

Residents located in proximity of major traffic corridors are exposed to particulate matter and diesel exhausts, causing a variety of respiratory and cardiovascular diseases [16]. Individuals residing in high-crime neighborhoods may indirectly develop stress-related behaviors, such as anxiety and higher blood pressure [17,18]. These instances illustrate the breadth of pathways through which both physical and social environments, as well as provision of health services, give rise to health disparities. Documenting the role of the geographic environment where individuals live and interact (often called “activity spaces”) will improve our understanding of health outcomes. This has deep policy implications for local health interventions and resource allocation decisions, ultimately leading to a reduction of health disparities.

A neighborhood is typically defined as the geographic area relevant to the specific health outcome being studied [8]. However, neighborhoods can be delineated by the extent of the individual's spatial interaction or administratively delimited at the scale at which policies are implemented (county, state). Secondary data sets—such as the American Community Survey published by the U.S. Census Bureau—help to overcome the absence of socioeconomic data in most U.S. medical records [19]. The scale at which the primary or secondary data sets are available often dictates the spatial granularity at which the analysis is conducted [20]. Spatial analyses with individual-level data from public health databases are also often limited by privacy policies required by data managers.

Patient information in the United States may be aggregated at the zip code level to comply with the Health Insurance Portability and Accountability Act [21]. As such, uniformity is assumed within the unit of analysis, but sharp contrasts may occur among adjacent units. Furthermore, the aggregation of neighborhoods does raise the issue of the modifiable areal unit problem; using different boundaries may lead to significantly different analytical results [22]. As a rule of thumb, analyses should be conducted at different scales to test the robustness of the spatial relationships and the effect of different artificial boundaries.

What defines a neighborhood and the concept of scale (aggregated, disaggregated) will influence the choice of methodologies and ultimately impact the results. Spatially based regression, contextual, and multilevel modeling are some of the key methods developed to incorporate neighborhood effects [8,23]. These

approaches allow researchers to estimate the impact of neighborhood effects after controlling for individual characteristics.

Role of geographic information systems in spatial epidemiology

Defined by Cromley and McLafferty [24] as “computer-based systems for the integration and analysis of geographic data,” geographic information systems (GIS) can describe, analyze, and predict patterns using feature (cartographic) and attribute data. GIS have been used in many epidemiologic applications, including disease mapping, rate smoothing, cluster or hot spot analysis, and spatial modeling. In its simplest form, GIS is often used to create spatially explicit variables such as availability and accessibility scores (e.g., food access), built environment measures (e.g., land use), environmental exposures (e.g., air pollutant concentrations), and demographic indicators (e.g., percent of persons in poverty). Measuring and describing the extent of spatial relationships is also a key function of GIS, which can be as simple as calculating the distance between two points or as complex as quantifying spatial dependencies in analytic models or identifying locally varying predictors. As described by Thornton et al. [25], GIS offers opportunities to integrate data across multiple databases and spatial scales for display, management, and analysis. In Table 1, we identify several ways to apply GIS tools to human immunodeficiency virus (HIV) research.

In descriptive epidemiology, thematic or color-shaded maps produced with GIS are useful for identifying areas at high risk for epidemics, highlighting population health disparities, examining resource needs, and ultimately, formulating hypotheses that lead to generation of explanatory models. Analytically, GIS tools can be used to explore spatial or spatiotemporal clustering, investigate locally varying relationships, and explicitly model or adjust for spatial dependencies in one's data.

Just as research has shown that interventions are most effective when implemented at multiple levels [33], we can gain more insight into individual-level outcomes by considering the context in which people live and work [8]. In the context of health care, it is equally important to consider the location and characteristics of where people seek their care (i.e., provider- and system-level factors). Contextual information can be considered in both descriptive and analytic ways. For example, one could explore the distribution of green space or recreational facilities within a region as well as develop a green-space index to predict the probability of meeting physical activity guidelines or individual weight status. When both

Table 1
Functions of GIS and related epidemiologic applications in HIV research

GIS functions	Application to HIV research
Store and measure spatial relationships	Distance between homes of newly diagnosed HIV cases and the closest Ryan White Clinic
Display spatial relationships	Bivariate map showing the relationship between county HIV prevalence and poverty rate
Analyze attribute and feature data simultaneously	Count of the number of HIV cases within 30 miles of each Ryan White Clinic
Manage data from multiple sources	Create a geodatabase with county HIV incidence and prevalence rates, Ryan White Clinic locations, community-based organizations providing HIV services, and county demographic indicators
Identify spatial patterns	Conduct a hot spot analysis of newly diagnosed HIV cases in U.S. counties
Explain spatial patterns	Determine where the prevalence of men who have sex with men is associated with the incidence rate for HIV in U.S. counties using GWR

A variety of studies have used GIS approaches in the context of HIV patient and service provider data [26–32].

individual- and area-level data are available, hierarchical models can be used to examine how outcomes vary at both the individual and/or group level (see example from Mobley et al., this issue [34]), as well as what factors outside an individual's immediate control impact their health and well-being. Generalized estimating equations, which also account for the correlation of persons within groups, treat this group-level effect as a nuisance that simply needs to be adjusted for in one's analysis. As epidemiologists, we need to be cautious about using methods to adjust away potentially meaningful information about the role of "place"—whether operationalized at the workplace, school, or residential level, on our health.

Study design perspectives

Cross-sectional studies typically examine the distribution of exposures and outcomes simultaneously, frequently through use of population-based administrative health data or representative sample surveys. Studies using cross-sectional designs rarely integrate the spatial perspective directly because spatial referents for each observation are not collected or are at a spatial resolution that does not permit localized investigations. For example, the National Health Interview Survey includes a data element for broad U.S. census regions (East, South, Midwest, West) [35], but nothing more spatially specific, whereas the public use version of the National Survey of Children's Health has a data element for state of residence and another for residence in a metropolitan statistical area [36], but these do not permit spatial analyses within a state. Similarly, surveys such as the Behavioral Risk Factor Surveillance System [37] will often mask county-level information due to small sample sizes. However, some states such as Florida routinely oversample to support zip code-level analyses every third year. Ecological studies are a form of cross-sectional study design in which the unit of analysis is grouped, by political unit (e.g., nation, state, county, zip code, census tract), health facility, school, or other organizational unit. Most ecological studies have the potential to incorporate the spatial perspective into their study designs and analyses. However, although researchers frequently map data from ecological studies using administrative or political units, the statistical analyses performed are typically aspatial (i.e., do not account for similarity or clustering between neighboring units). With tools such as spatial smoothing techniques, spatial regression, and multilevel modeling readily available to epidemiologists, more sophisticated analyses that take into account the spatial dependence present in many data sets should be considered.

Case-control study designs compare exposure histories of persons with a disease or health condition (cases) with persons who do not have that disease or health condition but are representative of the population from which cases are identified (controls) [1]. In general, the case-control study does not lend itself well to spatial epidemiologic analysis, unless cases and controls are selected from a population-based sampling frame that incorporates location into its design. However, there are several areas where the use of GIS tools and spatial statistical methods can enhance case-control studies. These include estimation of measures of access (e.g., distance to a health facility, travel times), neighborhood amenities, and other local estimates derived from spatial surfaces (e.g., food swamps and deserts, residential segregation, inequalities in income and wealth, or measures of neighborhood deprivation). If addresses are available for all cases and controls, these can be geocoded and linked to census level sociodemographic variables or environmental data (e.g., air pollutants, water quality, remotely sensed data on land use). For example, Lupo et al. found that neighborhood deprivation, measured at the census tract level, was associated with presence of cleft lip (with or without cleft palate) among Texas

residents in a birth defects registry [38]. If it is feasible to incorporate collection of spatially specific identifiers into case-control study designs, these studies may become more informative.

Cohort studies follow study subjects over time, comparing those with and without exposures of interest to determine their outcomes (disease, mortality) [1]. These study designs can take many forms, and use a variety of sophisticated biostatistical techniques. As with cross-sectional and case-control studies, epidemiologists typically do not incorporate spatial analysis directly into their study designs. Cohort study data can be integrated with census, environmental, and other administrative-level data when geocoded to specific addresses or administrative units (e.g., zip code centroids). More often than not, these studies are spatially enabled, in that the researcher uses spatial referents to acquire additional data to associate with each study subject, but do not incorporate spatial analysis into their study designs. A common use of GIS-derived data in cohort studies is the use of environmental data such as air pollution and other environmental contaminants on health outcomes such as mortality, cancer, or respiratory illnesses [39,40]. Additionally, studies have linked changes in the neighborhood food environment with body mass composition in prospective studies [41–43].

Methodologies

Geocoding and uncertainty

Locational information, for instance in the form of addresses, can be transformed into geographic coordinates through a process known as geocoding [44]. The procedure requires that addresses from the epidemiological dataset are standardized and comprised of an address number, street name, city or town name, state, and zip code [45]. Using a reference data set—typically a street database—geographic coordinates can be estimated by comparing and interpolating the address to the range of addresses for each segment of the reference data set. The procedure is sensitive to the completeness of the addresses such as the presence of "P.O. Box" and on the quality of local and regional street road network files [46,47].

The quality of the geocoding is evaluated based on the merits of three components [48]: the match rate (percentage of records being geocoded), the individual match score (how well the standardized address matches the street database), and the match type (e.g., geocoding at the street level or zip code). Low individual match score and low match type accuracy will increase positional errors, which are typically larger in rural areas. Such errors pose a serious challenge in spatial analysis because it may result in (1) an underestimation of local risk, (2) the misplacement of high-risk areas of a disease, (3) a misvaluation of spatial association, and (4) biased evidence for decision makers.

Geocoding can be implemented using either commercial GIS software or online. Although the setup costs to prepare the reference data set and standardize the addresses can be expensive when using commercial GIS software, this approach allows the researcher to geocode large amounts of records. Online geocoders are free and typically use more recent street network data, which is likely to result in lower positional errors if one's data are also fairly recent (if working with older data, the opposite is true). However, online geocoders are often limited as to how many records can be processed and raise important issues of confidentiality because addresses are uploaded online [48].

Because of privacy concerns such as Health Insurance Portability and Accountability Act, health departments may release geocoded address information only in aggregate form, such as for five-digit zip codes [24]. Alternatively, researchers can geomask the geocoded data to protect the privacy of individuals. Geomasking

modifies the geographic coordinates of an individual event by displacing the original location to some distance while maintaining spatial resolution for mapping purpose and cluster detection [49,50].

Distance estimation

Geographic accessibility to a health service depends on its availability and the impedance to access that service. Greater travel impedance will affect both the provision and utilization of health services [51]. A low level of geographic accessibility may have dramatic consequences in terms of health outcomes. Because of recent advances in GIS, different metrics can be used to estimate travel impedance such as travel distance (Euclidean or network-based), travel time, and travel cost [52,53]. Although estimating travel impedance by means of Euclidean distances is extremely straightforward, this approach can lead to serious underestimation of distances. When modeling travel, the ideal route between two points will be the one that minimizes the selected measure of impedance. Travel modeling should ideally incorporate speed limits, honor one-way restrictions, and reflect connectivity among roads—most of which can be specified in a GIS-based analysis. Travel time may be a more precise measure of impedance because it incorporates en-route conditions, such as congestion.

Online mapping providers such as Google Maps or MapQuest offer unprecedented support for estimating travel impedance. However, similar to geocoding services, the number of analytical queries that can be requested is usually limited, the accuracy of the travel estimates has not been extensively studied, and such approaches raise issues concerning data confidentiality. Through its Network Analyst extension, ArcGIS desktop software (ESRI, Redlands, CA, <http://www.esri.com/>) can produce distance and travel time estimates that account for road features with relative ease. Similarly, ArcGIS Pro can be used to calculate distance and travel time using ESRI's cloud-based road network data, although users are limited to no more than 1000 origin-destination pairs.

Although estimation of distance and/or travel time can now be done fairly simply, epidemiologists must consider which measures are most relevant to their population and/or region. The choice of origin and destination locations is also important, as variation in spatial scales, accuracy, and context can affect the results. For example, Khang et al. (this issue) [54] explores the probability of diagnostic resolution after an abnormal mammogram as a function of several operational definitions of distance: distance from the patient's residence to the screening facility, the diagnosing facility, and the closest mammography facility. Similarly, in a comparative analysis of GIS-based measures of access to mammography, Lian et al. [55] found that travel time and facility density were poorly correlated with odds of late-stage breast cancer, whereas spatial accessibility scores that considered both supply and demand (i.e., two-step floating catchment area) were significantly associated with late-stage diagnosis.

Residential mobility

Epidemiologists often include neighborhood factors in models examining determinants of individual health outcomes. In a GIS, neighborhood characteristics are assigned to individuals based on their residential location, a process known as spatial join. Results from this process are particularly informative to establish whether individual disease conditions may be caused by exposure to hazardous environment [56].

Although residential mobility is widely recognized in the research literature, typically only the attributes of the neighborhood where the individual resides at the time of diagnosis are

incorporated into spatial epidemiologic analyses [57]. Failure to include longitudinal residential profiles has serious implications because the location where the case was diagnosed may not necessarily coincide with the place where exposure started or occurred. In a seminal study by Pershagen et al. [58], the authors assessed the residential history of all study subjects, basing their exposure assessment on radon measurements at all dwellings in which the subject lived for 2 or more years over a period of 30+ years. The issue of residential histories is further exacerbated for (1) individuals with high mobility, (2) diseases with long periods of latency such as cancer, and (3) places where the conditions of the environment may change rapidly [59].

In maternal and child health, most studies assume that residential mobility does not occur during pregnancy; as such the use of maternal residence at birth may result in exposure misclassification, biased estimates of association, and imprecise risk estimation [60,61]. In a study by Miller [62] in the metropolitan Atlanta region, approximately 22% of pregnant women changed their residential location from the date of conception to delivery [61], although most relocation distances were short and residential location remained generally within the same county. Exposures that are homogenous within a community may be well estimated with limited residential data, but this assumption may not hold for longer distances.

Although the U.S. Census Bureau publishes migration information between counties and the proportion of residents who lived in the same residence 1 year ago [63], the lack of data at a more disaggregated level may be one reason why most epidemiological studies do not incorporate residential histories. Hughes and Pruitt (this issue) [64] suggest that electronic medical records derived address histories have the potential to alleviate this pitfall. More work is needed to develop spatiotemporal (cumulative) models of exposure [59].

Data linkages

Geocoding is a form of record linkage, in which address information is converted into latitude-longitude (X-Y) coordinates or administrative unit codes (e.g., Federal Information Processing Standard codes in the United States). For example, Barnes et al. (this issue) converted the self-reported address of each survey respondent to an X-Y coordinate for the purpose of measuring the presence, types of, and distance to food retailers around survey participants. More commonly, however, researchers link data from various databases across spatial units using geocodes. In a recent example [65], the association between income inequality and infant mortality was examined, using U.S. states as units of analysis. Although this study was aspatial, data from vital records were linked with socioeconomic data using state-level geocodes. Health services researchers and epidemiologists should be familiar with the Area Health Resources Files [66], providing hundreds of data fields with socioeconomic, demographic, and health services provider and utilization data by county. Similarly, epidemiologists frequently use geocodes for census units to access data from U.S. Census and associated American Community Survey databases. Increasingly, researchers also use geocodes to link individual records or census units surrounding their residences to environmental databases, using GIS technologies to integrate data on ambient air pollution, soil contamination, land use cover, temperature fluctuations, and other measures of phenomena in the physical environment with data on health behaviors and outcomes.

Overview of spatial and spatio-temporal clustering

According to McLafferty (2015) [67], a spatial cluster—also termed hot spot—can be defined as “...an unusual number of cases

within a population, place and time period ... ” Practically, an epidemiologist may be interested to determine whether disease rates around a hazard site are elevated or if the number of infected individuals is higher than would be expected. Accurate cluster information provides practical knowledge for public health interventions, such as screening, prevention, and surveillance. A series of robust statistical approaches have been developed to detect spatial or spatiotemporal clusters of diseases.

According to Walter Tobler’s First Law of Geography, “everything is related to everything else, but near things are more related than distant things [68].” This principle, otherwise known as spatial autocorrelation or dependence, is a key component of spatial epidemiology. Traditional generalized linear models that assume the independence of modeled observations will likely produce inaccurate estimates in the presence of spatial autocorrelation. It is generally desirable to estimate the amount of clustering using spatial search strategies, test its significance (for instance by means of Monte Carlo simulations), and finally visualize location of clusters along with their magnitude or extent. Global clustering tests evaluate the presence of clustering over the study area, whereas local clustering statistics tests for specific clusters at a finer scale. With the increasing technology and data available to incorporate contextual or locational data into our epidemiologic models, epidemiologists would be remiss to ignore the potential for spatial autocorrelation or nonstationarity in their data.

Spatial clustering

Evaluating the clustering of individual disease cases is typically implemented using a count statistic [69]. The K function is a clustering statistic which estimates the magnitude of clustering at different scales [70]. It uses a circular or elliptical search window over each case to count how many other cases occur in the near proximity. The window is then moved to the next points and the process repeated. The entire procedure is conducted at different search windows. The scale at which clustering is the greatest is then determined and can serve as an input for the kernel density estimation technique, which is essentially a point density function helping to visualize the location of clusters on the map [69]. Note that variable-size, adaptive filters can be used instead of static search strategies [67]; this is particularly important when there is a substantial amount of heterogeneity in the underlying population, such as the small numbers problem [71,72].

For clustering of cases aggregated into spatial units, spatial autocorrelation-based methods such as Moran’s I [73] tell us whether nearby units tend to exhibit similar rates. Spatial autocorrelation statistics measure the degree to which spatial features (e.g., zip codes, census tract) and their associated values tend to be clustered together in space or dispersed. Moran’s I value range from -1 to $+1$, with a value of -1 denoting that units with low rates are located near other units with high rates, whereas a Moran’s I value of $+1$ indicates a concentration of spatial units exhibiting similar rates. Although the Moran’s I statistic informs on the presence of clustering, it does not indicate where such clusters occur. The local version of the Moran’s I (local indicators of spatial association or LISA [74]) identifies both the locus and shape of these clusters. Another commonly used local clustering approach is the Kulldorff’s spatial scan statistic [75], which identifies the most likely disease clusters in a study area by maximizing the likelihood that disease cases are located within a set of concentric circles that are moved across the study area. Both number of observed and expected cases are counted, and the most “unusual” excess of observed cases is reported. The approach is flexible to incorporate heterogeneous background population densities and adjust for multiple testing. Several packages (i.e., R, GeoDa, ArcGIS, SaTScan)

can implement and visualize both global and local tests for clustering.

Space-time clustering

Several statistical approaches have been proposed to detect spatiotemporal clusters of diseases. For individual data, the Knox test for space-time interaction evaluates the presence of a space-time cluster at given spatial and temporal distances [76]. Knox’s method is limited due to its arbitrary definition of closeness, and the critical distance does not account for population heterogeneity [77,78]. The Mantel test [79] incorporates the notion of distance decay in that nearby pairs of events are more important than distant pairs. However, the Mantel statistic assumes a linear association between space and time, which is violated, especially for infectious diseases. Jacquez’s k -Nearest Neighbor k -NN statistic [78] addresses the weaknesses of the Knox and Mantel statistics, by counting the number of pairs of events that are nearest neighbors in both space and time. The space-time Ripley’s K function evaluates the magnitude of space-time clustering at different spatial and temporal scales [69]. Results from the Knox, Mantel, k -NN, or the space-time Ripley’s K function can serve as an input to visualize space-time patterns, for instance by means of the space-time kernel density estimation [80,81].

Local space-time clustering techniques include the space-time permutation statistic [82], which uses a cylinder with a circular (or elliptical) base, where the vertical axis represents time. Kulldorff’s space-time scan statistic [83] or the extension of the LISA statistic in time are generally used to estimate space-time clustering of aggregated data, such as space-time change in disease rates across a study region.

Although statistical tests of clustering are useful for retrospective analysis, prospective approaches such as the cumulative sum (cusum) method are designed for time series data which are updated on a regular basis, or when new events are observed [84]. These techniques are able to quickly detect deviations from the mean in a series of events to detect clusters as quickly as possible [85]. Software products such as SaTScan (<https://www.satscan.org/>) [86] and Geosurveillance (<https://www.acsu.buffalo.edu/~rogerson/geosurv.htm>) [84] incorporate prospective tests.

Small area estimation

Local health behavior and outcome data are important for program planning and evaluation, resource allocation, and policy-making activities. Although interest in local-level data is growing, population-based surveys powered to make national- or state-level inferences are rarely adequate to support direct estimation for small areas such as counties. Unit-level small area estimation techniques have become the optimal method to derive rates for counties and other small geographic areas given these known limitations. Indirect estimates for small areas are usually derived using synthetic or multilevel modeling. Although both approaches are based on prediction models, indirect estimates based on multilevel modeling also account for random area-level effects not explained by the covariates alone [87]. Multilevel model-based approaches to small area estimation have been extensively applied to population-based health surveys such as Behavioral Risk Factor Surveillance System on topics ranging from COPD, periodontitis, smoking prevalence, obesity, and HPV vaccine uptake [88–92]. A multilevel, post-stratification approach developed for polling research has been used in more recent years to provide granular and precise disease estimates at a variety of spatial scales. The post-stratification approach is unique in that it accounts for the underlying distribution of population such as age, race, and gender at the small area level (k) [91]. In essence, estimates are created for every possible

Table 2
Sample of data sources that permit geographic analysis of provider availability, treatment patterns, and health outcomes

Data source	Geographic levels	Description	Outside information
Healthcare Cost and Utilization Project (HCUP)	State, county, zip	Discharge-level information on inpatient, outpatient, and emergency department visits including charges, procedure and diagnosis codes, and demographics	http://www.hcup-us.ahrq.gov/
Medicare Fee-For-Service Provider Payment and Utilization Data	State, zip, address	Provider-level utilization and average payment information on services/procedures provided to Medicare beneficiaries	https://www.cms.gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/Medicare-Provider-Charge-Data/index.html
County Health Rankings	County	County-level rates of risk behaviors and health outcomes such as premature death, adult smoking, and teen births; online mapping tool available	http://www.countyhealthrankings.org/
Kids Count	County, region, congressional districts	Various demographic, economic, public safety, health behavior, and outcome data available	http://datacenter.kidscount.org/
Area Health Resources Files	County	County-level health status and provider availability indicators; online mapping tool available	http://ahrh.hrsa.gov/
RTI Spatial Impact Factors Dataset	County, ZCTA, PCSA, Tract, MSSA	Various health, transportation, environmental, and demographic indicators for multiple geographies	https://rtispacialdata.rti.org/
National Provider Identifier Registry	State, zip, address	Provider-level information including medical specialty	http://download.cms.gov/nppes/NPI_Files.html
Hospital Compare	State, zip, address	Hospital-level information including complications and readmissions, payment data, quality of care measures, and patient satisfaction scores	https://data.medicare.gov/data/hospital-compare
Medicare Geographic Variation Public Use File	State, HRR, county	Demographic, spending, utilization, and quality indicators for various geographic levels (some mapping tools available)	https://www.cms.gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/Medicare-Geographic-Variation/GV_PUF.html

HRR = hospital referral region; MSSA = medical service study area (California only); PCSA = primary care service area; ZCTA = zip code tabulation area.

subgroup combination and then averaged by their weighted distribution in small area *k*. In this issue, Lin et al. [93] apply this multilevel, post-stratification approach to estimation health-related quality of life in adults aged 65 years and older using data from the Behavioral Risk Factor Surveillance System. One can also use the spatial structure of the data to improve the small area estimates using the conditional autoregressive, intrinsic conditional autoregressive, or simultaneous autoregressive modeling frameworks.

Bayesian approaches to disease mapping

The primary focus of disease mapping is on identifying local variations in the incidence or prevalence of a health condition, disease, or health outcome. At its most elementary level, an analyst can plot the distribution of outcomes as a choropleth map, classifying each areal unit into one of a series of categories based on its relative frequency. Increasingly however, spatial epidemiologists are turning to Bayesian methods for disease mapping and small area estimation. These methods have become increasingly sophisticated as tools for simulation analyses now run very efficiently on desktop computers. Lawson provides a useful introduction to these approaches [4]. Two studies in this issue demonstrate how these methods are being extended to improve small area estimates of health outcomes (Neyens et al. [94]) and improve modeling of rare outcomes through spatiotemporal mixture modeling (Carroll et al. [95]).

Spatial regression models

Standard regression models such as the ordinary least squares (OLSs) capture the average strength and significance of the

predictor variables but make the assumption that rates at location *i* are independent of rates at neighboring units *j* and that the residuals are normally distributed. As such, OLS does not consider small-scale spatial variation and spatial autocorrelation in the residuals may indicate that those assumptions are violated [96]. There exist several techniques to incorporate spatial effects, such as the spatial regression [97] or the geographically weighted regression (GWR) technique [98]. In spatial regression, autocorrelation in the dependent variable—such as when rates in location *i* are influenced by rates at location *j*—justifies the use of a spatial lag model, which incorporates an autoregressive term for the dependent variable. Autocorrelation in the residuals, however, indicates that variables may have been omitted, and the prescription is to use a spatially lagged error model. Unlike OLS or spatial regression that provides global estimates, GWR is a local regression technique used to measure how the strength of the relationships among the dependent and explanatory variables differs from location to location. In essence, for each location, GWR pools data from adjacent neighborhoods to conduct a local regression, resulting in an estimate of local regression coefficients for each of the predictor variables.

Outlook

There is increasing interest by policy makers, public health practitioners, community planners, and researchers in understanding the social, environmental, and structural features of neighborhoods, which have been linked to a wide array of health outcomes. Identifying neighborhood-level assets and deficits provide a baseline for targeting resources and interventions appropriately to the unique needs of the underlying population. The stated goals of Healthy People 2020 [99], however,

demonstrate a move away from simply identifying health disparities, toward identifying and testing potential solutions that maximize each neighborhood's opportunity to achieve health equity.

At a foundational level, GIS software packages and online mapping applications offer a range of tools for stakeholders to visualize, analyze, and report on risk factors and outcomes at multiple geographic levels. These levels can have traditionally been defined based on administrative boundaries; however, empirical approaches based on the underlying data distribution rather than somewhat arbitrary geopolitical boundaries have also been proposed [100]. Using boundaries that mimic the actual travel patterns of the residents within areas or clustering of outcomes across areas seeks to minimize the modifiable area unit problem and understands how geography impacts areas differently. For example, when designing facility service areas or hospital markets, one can use patient referral data to draw hospital market boundaries [101] or use local provider competition data (e.g., number of medical providers around a locality) to draw variably sized market boundaries that account for the likelihood of travel outside the area [102].

Web-based mapping applications have also been on the rise, as technological advances such as application programming interfaces (APIs) including Google Maps API, ArcGIS Web APIs, and Bing Maps API, now readily enable programmers to build interactive Web-based mapping tools. On the user end, these applications (such as HealthLandscape [103], HealthMap [104], Community Commons [105], and Dartmouth Atlas [101]) allow for the creation of customized maps and reports, though rarely allow for the direct export of the underlying data. The vast majority of these online mapping applications have been restricted to the visualization of individual or aggregated data—partly due to protection of privacy and computational capabilities—but in recent years, several initiatives have been undertaken to incorporate spatial and spatio-temporal analytical capabilities over the Web [106–108], such as kernel density estimation and space-time linkages.

Fortunately, the era of big data and enhanced data sharing is here; epidemiologists working in many sectors will be affected and can positively impact the trajectory of the field. As evidenced by Vice President Joe Biden's Moonshot Cancer Initiative, enhanced data sharing is a major need for clinical and epidemiological research in the next decade. With small area (even address) level data being shared more readily with researchers (Table 2), more analyses are enabled that use a geospatial lens. Researchers looking to investigate geographic trends in provider availability, treatment patterns, and health outcomes may find resources such as www.healthindicators.gov offer a one-stop shop for access to geospatial data and is searchable by geographic level or risk factor and/or disease of interest. Many of the indicators listed on www.healthindicators.gov are cross-referenced to their respective Healthy People 2020 [99] objective as well. While www.healthindicators.gov will be retired in 2017, other tools such as www.healthdata.gov will continue to provide access to a variety of health databases.

With the growth in public data sharing and technologies to measure the neighborhood environment [109] (e.g., social media, crowdsourcing, Google Street View), epidemiologists have more tools at hand to monitor population health and identify actionable targets for intervention. Where possible, researchers must get away from measuring contextual influences based only on the residential environment, in static space and time. Life-course approaches, that consider the cumulative exposures people face as they age, make occupational shifts, and relocate over time are starting to surface, as are activity-space approaches that account for where people choose to engage in daily activities (see example

from Duncan et al., this issue [110]) [111,112]. With its inherent ability to integrate data from multiple systems, users, geographies, and time points, GIS holds great potential to help health researchers explore these challenges.

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